

Zhe (Barry) Chen

Distinguished Postdoctoral Fellow

Department of Building Environment and Energy Engineering

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WORK EXPERIENCE

2025.01–Present	The Hong Kong Polytechnic University Supervisor: Prof. Fu XIAO	Postdoctoral Fellow
2020.07–2021.07	University of Shanghai for Science and Technology Supervisor: Prof. Yongbao CHEN	Research Assistant

EDUCATION

2021.09–2024.12	The Hong Kong Polytechnic University Supervisor: Prof. Fu XIAO	PhD
2017.09–2020.04	Tongji University Supervisor: Prof. Peng XU	MEng
2013.09–2017.06	Nanjing University of Aeronautics and Astronautics	BEng

AWARDS & HONORS

2025	Champion (Ranking 1/610, AU\$10,000) , FlexTrack Challenge—Detecting Energy Flexibility in Buildings, AICrowd, New South Wales Government, and University of Wollongong
2025	FCE Awards for Outstanding PhD Theses , the Hong Kong Polytechnic University (Top 2.5%)
2024	Distinguished Postdoctoral Fellowship , the Hong Kong Polytechnic University; Finalist, RGC Junior Research Fellow Scheme (the only department nominee)
2024	Highly Cited Paper Award , <i>Advances in Applied Energy</i> (IF: 13.8) The most cited paper published in this journal in 2023
2023	Champion (CN¥100,000) , the First Great Bay Area Data Application Innovation Competition, the Development and Reform Commission of Shenzhen Municipality
2023	Excellent Control Award , the 21st MDV Central Air Conditioning Design and Application Competition, Midea Building Technologies
2023	Champion (Ranking 1/331, CN¥20,000) , the Schneider AI Master Cup Algorithm Competition, Schneider Electric and Baidu PaddlePaddle
2023	ESI Hot Paper and Highly Cited Paper , corresponding author paper titled “ <i>Physical energy and data-driven models in building energy prediction: A review</i> ”
2023	Champion (Ranking 1/513, CN¥20,000) , the UNiLAB Smart Energy System Algorithm Competition, UNiLAB and Alibaba Tianchi
2022	Paper nominated for the Eni Award 2023 , first author paper titled “Multi-objective residential load scheduling approach for demand response in smart grid”

2022 **Champion**, i-Space: VRAR Contest 2021/22, project titled “AR-enabled Smart Building Energy Management Platform”, the Hong Kong Polytechnic University
 2022 **Silver Award**, the Global AI Challenge for Building E&M Facilities, the Electrical and Mechanical Services Department
 2021 **Presidential PhD Fellowship**, the Hong Kong Polytechnic University

PUBLICATIONS

📄 Citations: 2,089; h-index: 18; hot paper: 1; highly cited paper: 2

Journal Papers

1. **Chen Z**, Xiao F*, Guo F. Similarity learning-based fault detection and diagnosis in building HVAC systems with limited labeled data. *Renewable and Sustainable Energy Reviews* 2023;185:113612. (IF: 16.3)
2. **Chen Z**, Xiao F*, Guo F, Yan J. Interpretable machine learning for building energy management: A state-of-the-art review. *Advances in Applied Energy* 2023;9:100123. (IF: 13.8, Highly Cited Paper Award)
3. **Chen Z**, Chen Y*, He R, Liu J, Gao M, Zhang L. Multi-objective residential load scheduling approach for demand response in smart grid. *Sustainable Cities and Society* 2022;76:103530. (IF: 12.0, Nominated for the Eni Award 2023)
4. **Chen Z**, Xiao F*, Chen Y. Multi-objective online optimization of building energy systems for improved control smoothness and efficiency. *Automation in Construction* 2026;181:106604. (IF: 11.5)
5. **Chen Z**, Zhang J, Xiao F*, Xu K, Chen Y. Development of a probabilistic cooling load prediction-based robust chiller sequencing strategy and its real-world implementation. *Applied Energy* 2025;382:125213. (IF: 11.0)
6. **Chen Z**, Xiao F*, Xiao Z, Chen Y. Bridging the gap between data-driven baselines and energy saving uncertainty for building retrofit. *Energy* 2025;340:139292. (IF: 9.4)
7. **Chen Z**, Zhang J, Xiao F*, Xu K, Chen Y. Physically consistent data-driven optimal sequencing strategy for variable speed pumps in large building chiller plants. *Building and Environment* 2025;281:113177. (IF: 7.6)
8. **Chen Z**, Chen Y*, Xiao T, Wang H, Hou P. A novel short-term load forecasting framework based on time-series clustering and early classification algorithm. *Energy and Buildings* 2021;251:111375. (IF: 7.1)
9. **Chen Z**, Xu P*, Feng F, Qiao Y, Luo W. Data mining algorithm and framework for identifying HVAC control strategies in large commercial buildings. *Building Simulation* 2021;14:63–74. (IF: 5.9)
10. **Chen Z**, Zhang J, Xiao F*, Madsen H, Xu K. Probabilistic machine learning for enhanced chiller sequencing: A risk-based control strategy. *Energy and Built Environment* 2025;6:783–95. (ESCI indexed, Scopus CiteScore: 17.3)
11. Chen Y, **Chen Z***. Short-term load forecasting for multiple buildings: A length sensitivity-based approach. *Energy Reports* 2022;8:14274–88. (IF: 5.1, Corresponding author)
12. Chen Y*, Guo M, Chen Z, **Chen Z***, Ji Y. Physical energy and data-driven models in building energy prediction: A review. *Energy Reports* 2022;8:2656–71. (IF: 5.1, Corresponding author, ESI Hot Paper, ESI Highly Cited Paper)

13. Tang H, **Chen Z**, Li H, Wang S*. Risk-aware optimal dispatch of resource aggregators integrating NGBost-based probabilistic renewable forecasting and bi-level building flexibility engagements. *Engineering* 2025;53:76–89.
14. Xu K, **Chen Z**, Xiao F*, Zhang J, Zhang H, Ma T. Semantic model-based large-scale deployment of AI-driven building management applications. *Automation in Construction* 2024;165:105579.
15. Chen Y, **Chen Z**, Xu P*, Li W, Sha H, Yang Z, et al. Quantification of electricity flexibility in demand response: Office building case study. *Energy* 2019;188:116054.
16. Chen Y, **Chen Z**, Ding Y, Du Y*, Li B, Liu J, et al. Towards Transformer Winding Deformation Fault Diagnosis under Data Scarcity: Integrating Frequency Response Analysis with Metric Learning. *IEEE Transactions on Dielectrics and Electrical Insulation* 2026:1–1.
17. Chen Y*, **Chen Z**, Yuan X, Su L, Li K*. Optimal control strategies for demand response in buildings under penetration of renewable energy. *Buildings* 2022;12:371.
18. Chen Y, Xu P*, **Chen Z**, Wang H, Sha H, Ji Y, et al. Experimental investigation of demand response potential of buildings: Combined passive thermal mass and active storage. *Applied Energy* 2020;280:115956.
19. Song Z, Xiao F*, **Chen Z**, Madsen H. Probabilistic ultra-short-term solar photovoltaic power forecasting using natural gradient boosting with attention-enhanced neural networks. *Energy and AI* 2025;20:100496.
20. Chen Y*, Yang Q*, **Chen Z**, Yan C, Zeng S, Dai M. Physics-informed neural networks for building thermal modeling and demand response control. *Building and Environment* 2023;234:110149.
21. Chen Y*, Wang H*, **Chen Z**. Sensitivity analysis of physical regularization in physics-informed neural networks (PINNs) of building thermal modeling. *Building and Environment* 2025;273:112693.
22. Xiao Z, Zhang J, **Chen Z**, Xiao F*. Probabilistic building energy savings verification via triplet network-based adaptive reference day selection. *Energy and Buildings* 2026;364:117664. (IF: 7.1)
23. Chen Y*, Chen Z, **Chen Z**, Yuan X. Dynamic modeling of solar-assisted ground source heat pump using Modelica. *Applied Thermal Engineering* 2021;196:117324.
24. Yuan H, Chen Y*, **Chen Z**. Investigation on Electricity Flexibility and Demand-Response Strategies for Grid-Interactive Buildings. *Buildings* 2025;15:4368.
25. Xiao T, Xu P*, Ding R, **Chen Z**. An interpretable method for identifying mislabeled commercial building based on temporal feature extraction and ensemble classifier. *Sustainable Cities and Society* 2022;78:103635.
26. Chu Y, Xu P*, Li M, **Chen Z**, Chen Z, Chen Y, et al. Short-term metropolitan-scale electric load forecasting based on load decomposition and ensemble algorithms. *Energy and Buildings* 2020;225:110343.
27. Xiao Z, Zhang J*, Xiao F*, **Chen Z**, Xu K, So PM, et al. An AI-enabled optimal control strategy utilizing dual-horizon load predictions for large building cooling systems and its cloud-based implementation. *Energy and Buildings* 2025;330:115352.
28. Yuan H, Chen Y*, Di Gangi A, **Chen Z**. Energy Flexibility Evaluation for Building Passive Thermal Storage Mass. *Energies* 2026;19:1035. (IF: 3.2)
29. Han C, Chen Y*, Wang H, Zhan S, **Chen Z**. Development and validation of a data quality

assessment framework for machine learning-based building load prediction. *Applied Energy* 2026;415:127925.

30. Sha H, Xu P*, Hu C, Li Z, Chen Y, **Chen Z**. A simplified HVAC energy prediction method based on degree-day. *Sustainable Cities and Society* 2019;51:101698.
31. Li A, Zhang J, Xiao F*, Fan C, Yu Y, **Chen Z**. Design information-assisted graph neural network for modeling central air conditioning systems. *Advanced Engineering Informatics* 2024;60:102379.
32. Gu J, Xu P*, Pang Z, Chen Y, Ji Y, **Chen Z**. Extracting typical occupancy data of different buildings from mobile positioning data. *Energy and Buildings* 2018;180:135–45.

Conference Papers and Presentations

1. **Chen Z**, Xiao F*. A novel demand response detection and capacity prediction framework. *Applied Energy Symposium: AIXEnergy*, May 2026, Ningbo, China.
2. **Chen Z**, Xiao F*. Reliably estimating energy savings using data-driven models as the baseline. *International Conference on Building and Simulation (BAS)*, July 2025, Borlänge, Sweden.
3. **Chen Z**, Xiao F*. Robust comparison of optimization algorithms for optimal chiller loading. *International Conference on Sustainable Development in Building and Environment (SuDBE)*, Aug. 2023, Espoo, Finland.
4. **Chen Z**, Xiao F*. Similarity learning-based FDD in building HVAC systems. *The 3rd International Conference for Global Chinese Academia on Energy and Built Environment International (CEBE)*, July 2023, Shanghai, China.
5. **Chen Z**, Chen Y*, Mohtaram S. Quantifying HVAC electrical flexibility from building thermal mass: a case study of the DOE reference building. *The 16th ROOMVENT Conference (ROOMVENT)*, Sept. 2022, Xi'an, China.
6. **Chen Z**, Xiao F*. Interpretable machine learning for AHU fault detection and diagnosis: Rule extraction from black-box models. *The 14th International Conference on Applied Energy (ICAE)*, Aug. 2022, Bochum, Germany.
7. **Chen Z**, Chen Y*. A multi-objective optimization-based load scheduling approach for demand response in smart grid. *The 2nd International Conference for Global Chinese Academia on Energy and Built Environment International (CEBE)*, July 2021, Chengdu, China.
8. **Chen Z**, Xu P*, Chen Y. A peer-to-peer electricity system and its simulation. *The 4th Asia Conference of International Building Performance Simulation Association (ASIM)*, Dec. 2018, Hong Kong, China.

Conference Session Chairs

1. *The 2nd International Conference on Digital Intelligence for Energy Systems (ICDIES)*, Oct. 2026, Wollongong, Australia.
2. *Applied Energy Symposium: AIXEnergy*, May 2026, Ningbo, China.

Invited Talks

1. Demand Response Detection and Capacity Prediction using Ensemble Learning. *IEEE International Conference on Energy Technologies for Future Grids*, Dec. 2025, Wollongong, Australia. (invited as the winner of FlexTrack Challenge)
2. Embarking on your research journey: My experience at PolyU BEEE. *Research Postgraduate*

Student Orientation, Department of Building Environment and Energy Engineering, The Hong Kong Polytechnic University, Sept. 2025, Hong Kong, China. (invited as PhD graduate representative)

PATENTS

1. A demand response control method for multi-energy residential buildings, CN202110226687.0.
2. A chiller control method, apparatus, equipment, and storage medium, CN118189539B.
3. An automatic generation method for schematic diagrams of HVAC water systems based on topological layered abstraction, CN202010507609.3.
4. A path optimization method for water systems in air conditioning systems, CN202011126801.4.
5. An optimization algorithm for pipe shafts, CN202011074593.8.
6. An automatic equipment selection algorithm for HVAC systems, CN202110515557.9.

RESEARCH PROJECTS

2025–2027	Development of Smart and Scalable Data-driven Management Technologies for Building HVAC Systems towards Carbon Neutrality PolyU Distinguished Postdoctoral Fellowship Scheme, HK\$841,040, P0053872 Role: Fellowship Awardee
2022–2024	Smart Data-Driven Building Management Framework for Environmental and Sustainability Applications Innovation and Technology Fund of Hong Kong, ITP/002/22LP Role: Lead researcher for demonstration of data-driven building management technologies in large buildings
2021–2024	Development of Smart Energy Management Technologies for Complex Building Energy Systems in High-Density Cities National Key Research and Development Program of China, 2021YFE0107400 Role: Lead researcher for data-driven modelling and optimization for building energy systems
2017–2021	Distributed Integrated Energy System Joint Simulation Technologies National Key Research and Development Program of China, 2017YFB0903404 Role: Lead researcher for distributed integrated energy system modeling

ACADEMIC SERVICE

Editorial Board	Young Editorial Board Member of <i>Carbon Neutrality</i> (IF: 12.5, Q1)
Guest Editor	Special issue of <i>Buildings</i> (ISSN 2075-5309) <i>Smart and Sustainable Buildings: Advancing towards Net-Zero and Intelligent Control</i>
Peer Reviewer	Invited reviewer for journals including: <i>Applied Energy</i> , <i>Automation in Construction</i> , <i>Energy Conversion and Management</i> , <i>Journal of Cleaner Production</i> , <i>Advanced Engineering Informatics</i> , <i>Engineering Applications of Artificial Intelligence</i> , <i>Computers in Industry</i> , <i>Energy, Building and Environment</i> , <i>Applied Thermal Engineering</i> , <i>Energy and Buildings</i> , <i>Building Simulation</i> , <i>iScience</i>
Membership	International Building Performance Simulation Association International Society of Energy and Built Environment

TEACHING EXPERIENCE

Advising	BSE3716 & BSE4726 (Capstone Research Project), 2025–2026	6 undergraduates
	BSE3713 & BSE4725 (Capstone Research Project), 2023–2024	2 undergraduates